

FHIR Telemedicine Workflow Architecture Guide

Overview

This document explains how to design a complete telemedicine workflow using FHIR.

This workflow supports:

- Online consultations
- Hybrid consultation models
- Offline fallback to OP workflow
- Patient-uploaded reports
- Doctor-uploaded reports
- AI-assisted workflows
- Video consultation workflows
- Prescription workflows
- Diagnostic workflows
- Follow-up workflows

This guide also explains:

- how DocumentReference should be used
- how historical reports should be handled
- how present consultation reports should be handled
- relationship between telemedicine and outpatient workflows

Core Workflow Principle

Your architecture should support:

Consultation Mode

- └─ online (telemedicine)
- └─ offline (outpatient/in-person)

IMPORTANT:

Telemedicine and outpatient should NOT be completely separate systems.

They should share:

- Patient

- Encounter
- Observation
- MedicationRequest
- DiagnosticReport
- Condition
- Invoice
- Claim
- Task

The main difference is:

Encounter delivery mode

Recommended Encounter Types

Use Encounter.class.

Telemedicine

```
"class": {  
  "code": "VR"  
}
```

Meaning:

Virtual encounter

Outpatient

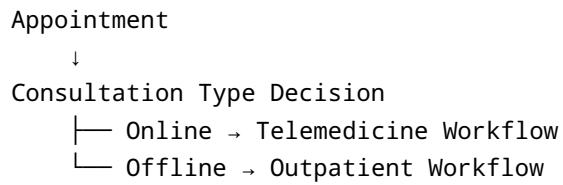
```
"class": {  
  "code": "AMB"  
}
```

Meaning:

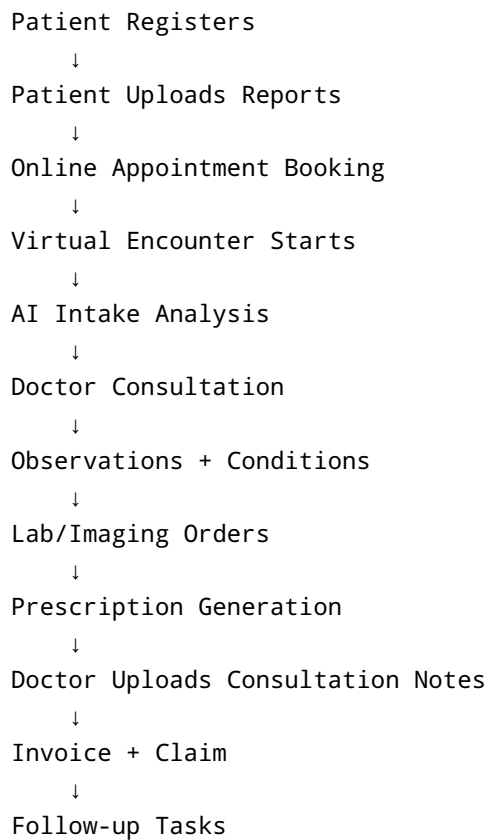
Ambulatory / in-person

HYBRID WORKFLOW MODEL

Recommended architecture:



TELEMEDICINE HIGH LEVEL FLOW



REQUIRED RESOURCES FOR TELEMEDICINE

Resource	Purpose
Patient	Patient record
Appointment	Online booking
Encounter	Virtual consultation
Practitioner	Doctor
QuestionnaireResponse	Intake forms
Observation	Clinical findings
Condition	Diagnoses
ServiceRequest	Lab/referral orders
DiagnosticReport	Test reports
MedicationRequest	Prescriptions
DocumentReference	Reports/files/documents
Invoice	Billing
Claim	Insurance
Task	Workflow orchestration

VERY IMPORTANT RESOURCE

DocumentReference

This is CRITICAL for telemedicine.

You should strongly implement it.

Without DocumentReference:

- report uploads become messy
- PDFs cannot be managed properly
- prescriptions cannot be attached
- scanned records become difficult
- AI document analysis becomes harder

What DocumentReference Stores

DocumentReference stores:

- PDFs
 - scanned reports
 - discharge summaries
 - prescriptions
 - radiology images
 - consultation notes
 - lab reports
 - ECG reports
 - insurance documents
 - external medical records
-

IMPORTANT QUESTION

Should reports be past or present?

ANSWER:

BOTH

You NEED both.

1. Historical Reports

These are uploaded BEFORE consultation.

Examples:

- old lab reports
- old prescriptions
- MRI scans
- discharge summaries
- historical ECGs
- previous diagnoses

Purpose:

Clinical context for doctor

Example Flow

```
graph TD
    A[Patient uploads old reports] --> B[DocumentReference created]
    B --> C[Doctor reviews before consultation]
```

Example Historical Report

```
{
  "resourceType": "DocumentReference",
  "id": "doc-old-report-1",
  "status": "current",
  "type": {
    "text": "Historical Lab Report"
  },
  "subject": {
    "reference": "Patient/patient-1"
  },
  "date": "2026-05-01T10:00:00+05:30",
  "description": "CBC report from previous hospital",
  "content": [
    {
      "attachment": {
        "contentType": "application/pdf",
        "url": "https://files.example.com/reports/cbc-old.pdf",
        "title": "CBC Report"
      }
    }
  ]
}
```

2. Current Consultation Documents

Generated DURING or AFTER consultation.

Examples:

- consultation summary
- e-prescription
- doctor notes
- AI summaries
- encounter summary
- diagnostic reports

Purpose:

Current episode documentation

Example Current Consultation Report

```
{
  "resourceType": "DocumentReference",
  "id": "doc-consult-summary-1",
  "status": "current",
  "type": {
    "text": "Telemedicine Consultation Summary"
  },
  "subject": {
    "reference": "Patient/patient-1"
  },
  "context": {
    "encounter": [
      {
        "reference": "Encounter/encounter-1"
      }
    ]
  },
  "date": "2026-05-17T14:00:00+05:30",
  "content": [
    {
      "attachment": {
        "contentType": "application/pdf",
        "url": "https://files.example.com/consult/summary.pdf",

```

```
    "title": "Consultation Summary"
  }
}
]
```

RECOMMENDED DOCUMENT CATEGORIES

Use categories.

Category	Example
historical-report	Old reports
consultation-note	Doctor notes
prescription	E-prescription
lab-report	Diagnostic reports
radiology-report	Imaging
discharge-summary	Hospital discharge
insurance-document	Insurance papers
consent-form	Telemedicine consent
ai-summary	AI-generated notes

TELEMEDICINE APPOINTMENT FLOW

STEP 1 — Patient Registration

Example

```
{
  "resourceType": "Patient",
  "id": "patient-1",
  "name": [
    {
      "text": "Rahul Sharma"
    }
  ]
}
```



```
]
}
```

STEP 2 — Upload Historical Reports

Patient uploads:

- old reports
- scans
- prescriptions
- images

Create:

```
DocumentReference
```

STEP 3 — Telemedicine Appointment Booking

Example

```
{
  "resourceType": "Appointment",
  "id": "appointment-1",
  "status": "booked",
  "serviceType": [
    {
      "text": "Telemedicine Consultation"
    }
  ],
  "participant": [
    {
      "actor": {
        "reference": "Patient/patient-1"
      },
      "status": "accepted"
    },
    {
      "actor": {
        "reference": "Practitioner/prac-1"
      },

```

```
    "status": "accepted"
  }
]
}
```

STEP 4 — Create Telemedicine Encounter

Example

```
{
  "resourceType": "Encounter",
  "id": "encounter-1",
  "status": "in-progress",
  "class": {
    "code": "VR"
  },
  "subject": {
    "reference": "Patient/patient-1"
  },
  "appointment": [
    {
      "reference": "Appointment/appointment-1"
    }
  ]
}
```

STEP 5 — Intake Questionnaire

Patient fills symptoms form.

Example

```
{
  "resourceType": "QuestionnaireResponse",
  "id": "qr-1",
  "status": "completed",
  "subject": {
    "reference": "Patient/patient-1"
  },
}
```

```
"item": [  
  {  
    "text": "Symptoms",  
    "answer": [  
      {  
        "valueString": "Fever and cough"  
      }  
    ]  
  }  
]
```

STEP 6 — AI Intake Analysis (Optional)

AI agent analyzes:

- symptoms
- uploaded reports
- historical conditions

Outputs:

- urgency score
- suggested specialty
- doctor briefing

STEP 7 — Doctor Consultation

Doctor reviews:

- uploaded historical reports
- live symptoms
- questionnaire
- previous encounters

Doctor records:

- Observation
- Condition

Observation Example

```
{
  "resourceType": "Observation",
  "id": "obs-1",
  "status": "final",
  "code": {
    "text": "Reported Temperature"
  },
  "subject": {
    "reference": "Patient/patient-1"
  },
  "encounter": {
    "reference": "Encounter/encounter-1"
  },
  "valueString": "100 F"
}
```

Condition Example

```
{
  "resourceType": "Condition",
  "id": "condition-1",
  "clinicalStatus": {
    "text": "active"
  },
  "code": {
    "text": "Upper Respiratory Infection"
  },
  "subject": {
    "reference": "Patient/patient-1"
  },
  "encounter": {
    "reference": "Encounter/encounter-1"
  }
}
```

STEP 8 — Lab Orders

Doctor creates ServiceRequest.

Example

```
{
  "resourceType": "ServiceRequest",
  "id": "sr-cbc-1",
  "status": "active",
  "intent": "order",
  "code": {
    "text": "Complete Blood Count"
  },
  "subject": {
    "reference": "Patient/patient-1"
  },
  "encounter": {
    "reference": "Encounter/encounter-1"
  }
}
```

STEP 9 — Prescription Generation

MedicationRequest Example

```
{
  "resourceType": "MedicationRequest",
  "id": "medreq-1",
  "status": "active",
  "intent": "order",
  "subject": {
    "reference": "Patient/patient-1"
  },
  "encounter": {
    "reference": "Encounter/encounter-1"
  },
  "medicationCodeableConcept": {
    "text": "Paracetamol 500mg"
  }
}
```

STEP 10 — Generate E-Prescription PDF

Strong recommendation:

Generate PDF prescription.

Store as:

DocumentReference

Prescription Document Example

```
{
  "resourceType": "DocumentReference",
  "id": "doc-prescription-1",
  "status": "current",
  "type": {
    "text": "E-Prescription"
  },
  "subject": {
    "reference": "Patient/patient-1"
  },
  "context": {
    "encounter": [
      {
        "reference": "Encounter/encounter-1"
      }
    ]
  },
  "content": [
    {
      "attachment": {
        "contentType": "application/pdf",
        "url": "https://files.example.com/prescriptions/rx-1.pdf"
      }
    }
  ]
}
```

STEP 11 — Doctor Consultation Notes

Strong recommendation:

Doctor notes should also be stored as:

DocumentReference

STEP 12 — AI Summary Generation

AI can generate:

- encounter summary
- patient-friendly summary
- follow-up instructions

Store using:

DocumentReference

STEP 13 — Billing

Generate:

- Invoice
- Claim
- ClaimResponse

Same as outpatient workflow.

STEP 14 — Follow-up Workflow

Doctor may recommend:

- another telemedicine visit
- physical OP visit
- emergency escalation

- lab follow-up
-

IMPORTANT HYBRID WORKFLOW

Online → Offline Transition

Very important pattern.

Example:

```
Telemedicine consultation
↓
Doctor suspects serious issue
↓
Create OP Appointment
↓
Physical consultation required
```

Example Workflow

```
Virtual Encounter
↓
Doctor detects chest pain risk
↓
Task: Emergency OP Evaluation
↓
Create In-Person Appointment
↓
Follow Outpatient Workflow
```

IMPORTANT DESIGN PRINCIPLE

Telemedicine should NOT duplicate outpatient.

Instead:

Telemedicine = Another Encounter Mode

Same resources.

Different interaction model.

RECOMMENDED TELEMEDICINE-SPECIFIC FIELDS

Appointment

Add extensions or custom fields:

Field	Purpose
consultationMode	online/offline
videoMeetingUrl	video link
platform	Zoom/Meet/WebRTC
waitingRoomStatus	queue handling

Encounter

Add:

Field	Purpose
virtualPlatform	platform used
recordingAvailable	recording metadata
telemedicineConsent	consent tracking

DOCUMENT STORAGE ARCHITECTURE

Recommended Pattern

FHIR stores metadata
Actual files stored in object storage

Examples:

- S3
- MinIO
- Cloudflare R2
- GCS

Recommended Document Flow

Upload File
↓
Store in Object Storage
↓
Generate Secure URL
↓
Create DocumentReference

RECOMMENDED SECURITY

VERY IMPORTANT for telemedicine.

Protect:

- uploaded reports
- prescriptions
- video session metadata
- AI summaries
- consultation notes

Recommended Security Features

Feature	Purpose
signed URLs	secure document access
expiring links	prevent leaks
audit logging	compliance
encryption	privacy
RBAC	access control

AI WORKFLOWS FOR TELEMEDICINE

Very powerful area.

AI Intake Agent

Analyzes:

- uploaded reports
- symptoms
- historical conditions

Outputs:

- triage
 - urgency
 - doctor summary
-

AI Report Summarization

AI reads uploaded PDFs.

Generates:

- structured findings
- trend summaries
- alerts

AI Follow-Up Agent

Example:

```
Lab abnormality detected
↓
AI creates follow-up task
↓
Patient notified
```

TELEMEDICINE TASK WORKFLOW

Example

```
Appointment Booked
↓
Task: Verify Uploaded Reports
↓
Task: AI Intake Summary
↓
Task: Doctor Consultation
↓
Task: Generate Prescription PDF
↓
Task: Billing
↓
Task: Follow-up Reminder
```

RECOMMENDED ADDITIONAL RESOURCES

Strong recommendations for telemedicine:

Resource	Why Needed
DocumentReference	CRITICAL
Binary	Direct file storage support

Resource	Why Needed
Consent	Telemedicine consent
Provenance	AI/doctor audit trail
AuditEvent	Compliance
AllergyIntolerance	Medication safety
Medication	Prescription catalog
Task	Workflow orchestration

RECOMMENDED PRODUCTION RESOURCE SET

Patient
 Appointment
 Encounter
 Practitioner
 Observation
 Condition
 ServiceRequest
 DiagnosticReport
 MedicationRequest
 DocumentReference
 Task
 Invoice
 Claim
 ClaimResponse
 QuestionnaireResponse
 Consent
 AllergyIntolerance

FINAL ARCHITECTURE PRINCIPLE

Use:

Telemedicine = Virtual Encounter Layer
 Outpatient = Physical Encounter Layer
 FHIR = Unified Healthcare Data Model

Both should share:

- workflows
- resources
- billing
- claims
- AI orchestration
- patient history

Only consultation mode changes.

FINAL SUMMARY

YES — telemedicine absolutely needs:

DocumentReference

And YES — it should support:

Past reports + Current reports

Because doctors need:

- historical context
- current consultation documents
- uploaded diagnostics
- prescriptions
- AI summaries
- follow-up reports

The best architecture is:

```
Single Unified FHIR Platform
├── Online Telemedicine Workflow
└── Offline Outpatient Workflow
```

This architecture scales very well and avoids duplicate systems.